

Ben Vaccaro

Boston, MA

Radar sensing · machine learning systems · neural interfaces

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EDUCATION

MIT Beaver Works Summer Institute — UAS-SAR

Summer 2026

MIT Lincoln Laboratory

- Synthetic-aperture radar on unmanned aircraft. Admitted as 1 of 21 from 400+ applicants.

Millis High School

2023–2027

Millis, MA · Class of 2027 · 95/100

- AP Physics C (Mechanics, E&M), Calculus AB, Statistics, Biology, Computer Science Principles, Literature, Language, Spanish, US History.
- Dual enrollment in engineering, University of Texas at Austin.
- VP Student Council (9–10, 12); National Honor Society (11–12); Peer Leadership Club, Student IT Volunteer (9–12).

RESEARCH & EXPERIENCE

Research Assistant — UAS-SAR

2026–

MIT Lincoln Laboratory, Beaver Works

- Synthetic-aperture radar carried on drones: signal processing and flight-side capture.

Founder, Rev A

2026–

Solderable

- Wireless brain–computer interface platform, funded by a Rev A hardware grant.

AI Research

2026–

Mantis AI, MIT CSAIL

- Training-data pipelines for AI systems engineering.

Software Engineer

2026–

Synthica

- Research-publication infrastructure: the software and deployment layer carrying scholarly work from draft to reader.

Team Captain

2024–

SeaPerch Underwater Robotics

- Underwater ROV design and competition; top 50 nationally in the team’s first year.

Stardance Ambassador

2025–

Hack Club × NASA

- Hackathon organizing and community outreach.

PUBLICATIONS

- First paper in preparation — details to follow.

HONORS & AWARDS

- **1st overall**, MIT Blueprint — winning prototype built in ten hours (2026).
- **4th overall**, Special Competitive Studies Project at Harvard (2026).
- **Xerox Innovation and Technology Award** (2026).
- **MIT LLRISE** — accepted, declined for BWSI (2026).
- **\$10k directed to local nonprofits** as YIP grant lead, Community Foundation for MetroWest (2025–2026).

TECHNICAL

Signals	Synthetic-aperture radar, remote sensing, DSP
Hardware	PCB design and bring-up, embedded firmware, wireless BCI
Software	Python, C/C++, JavaScript, Git, Linux, CI/CD
ML	Training-data pipelines, AI systems engineering